6	**(a)**	**(i)**	coherence: constant phase difference	M1	
			between (two) waves	A1	[2]
		(ii)	path difference is *either* λ *or* $n\lambda$		
			or phase difference is 360° *or* $n \times 360°$ *or* $n2\pi$ rad	B1	[1]

(iii)	path difference is *either* $\lambda/2$ *or* $(n + ½)\ \lambda$ *or* phase difference is odd multiple of *either* 180° *or* π rad	B1	[1]
(iv)	$w = \lambda D / a$	C1	
	$= [630 \times 10^{-9} \times 1.5] / 0.45 \times 10^{-3}$	C1	
	$= 2.1 \times 10^{-3}$ m	A1	[3]
(b)	no change to <u>dark</u> fringes	B1	
	no change to separation/fringe width	B1	
	<u>bright</u> fringes are brighter/lighter/more intense	B1	[3]
7 (a) (i)	2 protons and 2 neutrons	B1	[1]